

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

7tLJBQPRtdh3UcG1Ecy2G3B1eSi51m4DciKVovaE8Yvz

Date: 5/1/2025

Compounds: Adderall, Codeine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Addine

Formula: C₂₅H₃₄N₃O₃

Weight: 424.56

Drug-likeness: 0.75

Activity:

CNS Stimulant and Analgesic

Target Analysis

Monoamine Transporters (DAT, NET, SERT)

Binding Affinity: 8.5

Modulation of mood and alertness through neurotransmitter uptake inhibition.

Mu-opioid Receptor

Binding Affinity: 7.2

Reduction of pain perception through opioid receptor agonism.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Cardiovascular system, Central nervous system, Liver

Mitigation Suggestions:

- Regular monitoring of cardiovascular function
- Use only under strict medical supervision

Pharmacokinetics

Absorption:

Rapid, oral bioavailability around 70%

Distribution:

Widely distributed across blood-brain barrier

Metabolism:

Primarily hepatic via CYP2D6 and CYP3A4

Excretion:

Renal, primarily as metabolites

Half-life: 6-10 hours

Detailed Analysis

The proposed compound Addine represents a novel approach by simultaneously targeting neurotransmitter transporters and the opioid receptors, thereby combining the CNS stimulant effect of Adderall-like compounds with the analgesic qualities of codeine analogs. The design ensures appropriate binding affinity to crucial transporters like DAT, NET, and SERT, providing stimulant effects, while the engagement with the mu-opioid receptors offers pain relief. While it offers therapeutic potential for dual conditions especially where augmentation of stimulant and analgesic therapies is warranted, its use poses risks typical of both stimulant and opiate drugs, necessitating cautious deployment with strict monitoring protocols.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>